Clinical EEG and Neuroscience
2024, Vol. 55(5) 543–552

DOI: 10.1177/15500594231222980
journals.sagepub.com/home/eeg
Sage

Application of Hybrid DeepLearning Architectures for Identification of Individuals with Obsessive Compulsive Disorder Based on EEG Data

Shams Farhad[1], Sinem Zeynep Metin[2], Çağlar Uyulan[3], Sahar Taghi Zadeh Makouei[4], Barış Metin[5], Türker Tekin Ergüzel[6], and Nevzat Tarhan[2]

Abstract

Objective: Obsessive-compulsive disorder (OCD) is a highly common psychiatric disorder. The symptoms of this condition overlap and co-occur with those of other psychiatric illnesses, making diagnosis difficult. The availability of biomarkers could be useful for aiding in diagnosis, although prior neuroimaging studies were unable to provide such biomarkers. **Method:** In this study, patients with OCD were classified from healthy controls using 2 different hybrid deep learning models: one-dimensional convolutional neural networks (1DCNN) together with long-short term memory (LSTM) and gradient recurrent units (GRU), respectively. **Results:** Both models exhibited exceptional classification accuracies in cross-validation and external validation phases. The mean classification accuracies in the cross-validation stage were 90.88% and 85.91% for the 1DCNN-LSTM and 1DCNN-GRU models, respectively. The inferior frontal, temporal, and occipital electrodes were predominant in providing discriminative features. **Conclusion:** Our findings underscore the potential of hybrid deep learning architectures utilizing EEG data to effectively differentiate patients with OCD from healthy controls. This promising approach holds implications for advancing clinical decision-making by offering valuable insights into diagnostic markers for OCD.

Keywords

obsessive-compulsive disorder, deep learning, long-short term memory, gradient recurrent unit

Received August 10, 2023; revised December 6, 2023; accepted December 8, 2023.

Introduction

Obsessive-compulsive disorder (OCD) is a chronic condition characterized by frequent intrusive thoughts (obsessions) and an urge to perform repeated behaviors (compulsions) to diminish the discomfort associated with obsessions. The lifetime incidence rate of OCD is estimated to be around 2%.[1] On the other hand, 28% of the population reports having obsessions or compulsions at some time during their life.[1] In addition, obsessive-compulsive personality disorder is very common (6.5%), and highly comorbid with OCD, and its clinical presentation may be very similar to OCD.[2] Finally, OCD was once classified as an anxiety disorder and the symptoms quite often overlap with anxiety, or individuals with anxiety may have accompanying obsessions and compulsions.[3] The comorbid and overlapping conditions may create diagnostic confusion and delay clinical practice. Therefore biomarkers may be useful in identifying individuals with OCD.

Neuroimaging methods can be quite useful to serve as biomarkers. Structural neuroimaging studies reported thinning in temporoparietal cortices in OCD[4] however integration of those differences into machine learning algorithms did not reveal useful biomarkers.[5] The authors concluded that this lack of successful classification may be due to medication effects.[5] Regarding functional magnetic resonance imaging (fMRI) studies, Zhang et al[6] validated a monkey-derived OCD classifier in a human dataset using resting state fMRI and reported 78%

[1]Department of Neuroscience, Uskudar University, Istanbul, Turkey
[2]Department of Psychiatry, Uskudar University, Istanbul, Turkey
[3]Department of Mechanical Engineering, İzmir Katip Çelebi University, İzmir, Turkey
[4]Faculty of AI engineering, Institute of Science, Uskudar University, Istanbul, Turkey
[5]Medical Faculty, Neurology Department, Uskudar University, Istanbul, Turkey
[6]Faculty of Engineering and Natural Sciences, Department of Software Engineering, Uskudar University, Istanbul, Turkey

Corresponding Author:
Barış Metin, Saray, Ahmet Tevfik İleri Cd No:18, 34768 Ümraniye/İstanbul, Turkey.
Email: baris.metin@uskudar.edu.tr

Full-color figures are available online at http://journals.sagepub.com/home/eeg.

accuracy. The regions that differentiated the clinical sample from controls included prefrontal, somatosensory, temporal, and parietal nodes.[6] Apart from functional and structural magnetic resonance imaging, electroencephalography (EEG) can also be used to derive biomarkers.

A study measuring phase-lag connectivity between local field potentials and EEGs revealed that connectivity between mid-frontal and ventral regions is associated with OCD symptom severity and treatment outcomes, suggesting frontostriatal connectivity as a symptom-related biomarker of OCD.[7] In another study transcranial-direct current stimulation (tDCS) stimulation of the right orbitofrontal cortex was found to reduce both TMS-evoked N100 amplitude and OCD symptoms, reinforcing the significance of frontostriatal brain regions as indicative of OCD.[8]

EEG biomarkers of OCD have been effectively utilized in the classification of the disorder using artificial intelligence (AI) tools. For instance, previously we employed machine learning (ML) algorithms using EEG band power measures to predict the response to transcranial magnetic stimulation in OCD, achieving 80% accuracy with the theta band.[9] In addition, we applied an ML algorithm to detect treatment resistance based on EEG complexity in OCD and obtained 90% accuracy.[10] Nevertheless, based on the paucity of biomarker studies in OCD, one might conclude that more studies are needed to detect accurate biomarkers for diagnosis, prediction of treatment resistance, and response. Deep learning algorithms could transform and improve the clinical applications for EEG-based detection of neurodegenerative diseases.

Recent studies have a common application of deep learning-based automatic identification systems using EEG signals.[11–21] A study by Balcı and Oralhan[11] used the long-short term memory (LSTM) algorithm for an EEG-based identification system, which has achieved an average accuracy rate of 86.29%. In another study, researchers successfully classified epilepsy disease and epileptic seizure moments across 5 distinct classes, including healthy groups, interictal cases, and seizure episodes, using a one-dimensional convolutional neural network (1DCNN) deep learning architecture, with accuracies as high as 100% for binary healthy-epilepsy comparison, and 94.17% for classifying patients into all 5 categories.[12] Moreover, a variety of deep learning architectures, including 1DCNN, 2DCNN, LSTM, 1DCNN-LSTM, and 2DCNN-LSTM, were implemented to differentiate patients with major depressive disorder from healthy subjects. The 1DCNN-LSTM model, which attained a classification accuracy of 99.24% showed the highest performance among these models.[13] A comparative study used 2 separate deep learning architectures, one based on a 1DCNN model and the other using a 1DCNN-LSTM architecture, to classify patients with unipolar depression and healthy controls. The models reached classification accuracies of 98.32% and 96.97%, respectively.[14] Also, Xu et al[15] suggested a 1DCNN-LSTM technique for automatically identifying epileptic seizures through EEG data. Their method reached high accuracies, with 99.39% accuracy for the binary epileptic seizure classification and 82.00% accuracy for the more difficult 5-class epileptic seizure recognition.[15] Also, a study implemented a customized AlexNet network reaching 90% accuracy in diagnosing autism spectrum disorder.[22]

Deep learning techniques have also been effectively used in a variety of fields. These include decoding motor imagery,[16] examining spatiotemporal characteristics in EEG data,[17] determining sleep stages,[18] comprehending emotions,[19] as well as recognizing actions[20] and detecting targets.[21] The classification accuracy recorded in these investigations was high, ranging from 76.6% to an excellent 99.98%.

In this study, we used deep learning for the first time using EEG signals to identify biomarker features in individuals with OCD and to differentiate them from healthy controls. This study presents 1DCNN-LSTM and 1DCNN-GRU (gradient recurrent units) models, which can directly process the original EEG signals collected from OCD and healthy subjects, and extract the features that are related to the OCD diagnosis.

Materials and Methods

Participants

This study included 52 healthy subjects (30 males and 22 females, mean age ± SD: 31.731 ± 10.347 years ranging from 18 to 58) and 86 patients with OCD (42 males and 44 females, mean age ± SD = 31.302 ± 10.009 years ranging from 18 to 60). Age and sex differences between the 2 groups were insignificant (Table 1).

The clinical data of the subjects were gathered retroactively from the database of the Hospital. The experimental group consisted of patients with OCD diagnosis. The control group consists of patients who applied for a neuropsychiatric checkup and in whom no major pathology was observed during neuroimaging tests and neuropsychological assessment. The control group, in contrast, consisted of individuals who underwent a neuropsychiatric checkup and in whom no significant pathology was discovered during neuroimaging and cognitive evaluation. Brain injuries, neurological conditions, and substance use at the time of data collection were all exclusion factors.

EEG Data Acquisition

Retrospective EEG data were originally recorded in the eyes-closed condition for a total of 3 minutes. The EEG

Table 1. Demographic Characteristics of Participants.

	Groups OCD (n = 86)	Control (n = 52)	Significance**
Gender	42 M, 44 F	30 M, 22 F	0.313
Age*	31.302 (±10.009, 18-60)	31.731 (±10.347, 18-58)	0.810

*Age values are presented in years as mean (standard deviation and range).
**No significant gender ($\chi^2 = 1.018$; $P = .313$) and age ($t = 0.241$; $P = .810$) differences between groups.
Abbreviations: F, female; M, male; *n*, number of subjects in each group; OCD, obsessive-compulsive disorder.

recording protocol was the same as reported in our previous project.[23]

International 10 of 20 nomenclature was used to record 19-channel EEG with active electrodes embedded in a cap (Wuhan Greentek Pty. Ltd, Wuhan, China). The electrode impedance was maintained below 5 KΩ. Using Scan LT (version 1.2), EEG signals were captured at 125 Hz with the ground electrode positioned halfway between the Fpz and FZ electrodes.

EEG Data Preprocessing and Analysis

The data were first filtered from 0.1 to 50 Hz. Using the Brain Vision Analyzer Software (version 2.2), Independent Component Analysis (ICA) was performed on the data to eliminate blink-related contaminations and eye movement artifacts in the EEG segments. Following ICA, the data was checked by eye and cleaned from the remaining problematic fragments. Then, for additional processing, data files were exported .edf files.

Deep Learning Architectures

Convolutional neural networks (CNNs) serve as powerful tools in automatically extracting features, primarily excelling in tasks like image classification due to their adeptness at capturing spatial patterns. They consist of 4 types of layers: convolutional, activation, pooling, and fully connected. The convolutional layer is in charge of utilizing filters to extract features from the input data. The activation function adds nonlinearity to the network and the pooling layer reduces the size of the output from the convolutional layer to reduce computational effort and prevent overfitting. The fully connected layer functions like a traditional neural network but has a large number of parameters.

When it comes to time series modeling, recurrent neural network (RNN) is one of the core deep learning approaches. Since EEG data consist of amplitudes in the temporal domain, the waveform is an ideal fit for RNN analysis. However, the method is improved by utilizing LSTM and GRU networks with an augmented forget gate and some dropout regularization due to the vanishing gradient problem of RNN. Being a special type of RNN, LSTM architecture has feedback connections that distinguish it from the standard neural networks. It remembers values at random times with feedback networks and eliminates problems such as long-forgetting caused by RNN.

Another property of LSTM networks is the ability to add or remove information from cell states through gates. LSTM deep learning unit consists of 3 gates that are classified as the forget gate, the entrance gate, and the exit gate, respectively. The forget gate consists of the sigmoid layer and decides what information to discard. The entrance gate consists of sigmoid and hyperbolic tangent layers. While the sigmoid layer decides what information to update, the tangent layer creates a vector of updated new values. The exit gate consists of sigmoid and hyperbolic tangent layers. First, updated cell state information is passed through the hyperbolic tangent layer. Next, the current input information and the output of the previous cell are passed through the sigmoid layer. Finally, the information from both layers is multiplied to obtain the cell output expression. Because LSTM can learn long-term dependencies, it addresses the long-term dependency issues that arise in RNNs and solves the problem of vanishing gradients in time series analysis. Long-term forgetting issues are unknown time delays between events found in time series analysis.

GRU is an architecture that deals with the vanishing gradient problem that arises in RNNs. Like LSTM, GRU may also learn long-term dependencies. LSTM has an exit gate, whereas GRU does not, despite having a forget gate. Therefore, it has fewer parameters. Because GRU uses fewer parameters, it performs better on smaller and more sparse datasets. Reset and update gates are 2 of the GRU's ports. The enter-and-forget port of the LSTM is replaced by the update gate. Additionally, the GRU lacks a distinct memory cell. In short, the main purpose of GRU and LSTM is to identify and track what information in the past can be kept and forgotten. The simplified representations of the LSTM and GRU structures are given in Figure 1.

To sum up, deep learning is an AI technology with powerful algorithms that deduplicate feature extraction, size reduction, and classification steps. The CNN is capable of automatically extracting the features. It applies primarily to the 2-dimensional convolutional neural network (2DCNN), which is suitable for image classification. In our application, the EEG signal is 1-dimensional waveform data, thus, "1DCNN" was utilized for the automatic feature extraction of the EEG signal.

In this study, classification performances were compared by using the 1-dimensional convolutional neural network (1DCNN) together with LSTM and GRU architectures, respectively.

The rationale behind using 1DCNN is to exploit its capability in automatic feature extraction from 1-dimensional EEG waveform data. Due to the critical importance of identifying long-term patterns and dependencies within EEG data, LSTM, and GRU architectures are selected over traditional RNNs due to their heightened proficiency in addressing challenges associated with gradient vanishing or exploding during training, a common issue encountered in basic RNNs.

While LSTMs excel in handling complex sequences due to their sophisticated memory mechanisms, they also come with increased computational complexity. Conversely, GRUs, with their simplified structure, offer computational efficiency, particularly evident in faster training times, especially on smaller datasets. Thus, the choice of these hybrid architectures stemmed from a consideration of the trade-offs between the complexity and efficiency of the GRU and LSTM networks.

Hybrid Architectures

We propose a neural network architecture for a binary classification task, illustrated in Figure 2. Firstly, TensorFlow Keras

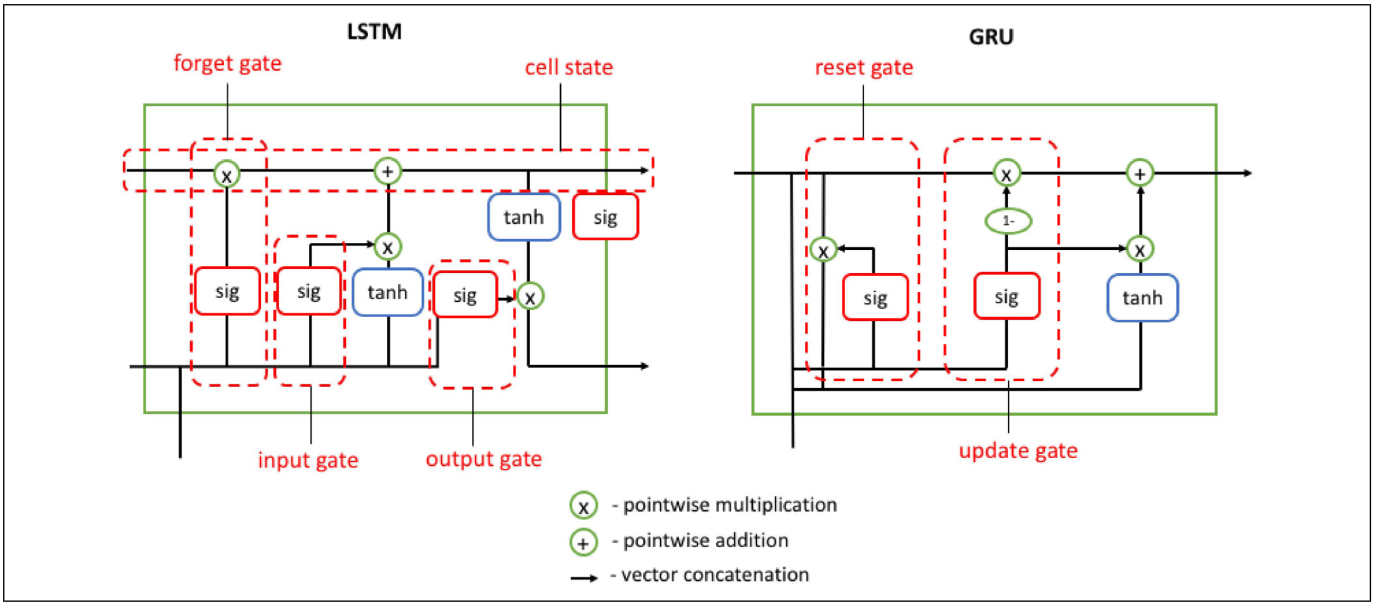

Figure 1. The simplified representation of the LSTM and GRU structures.
Abbreviations: GRU, gradient recurrent units; LSTM, long-short term memory.

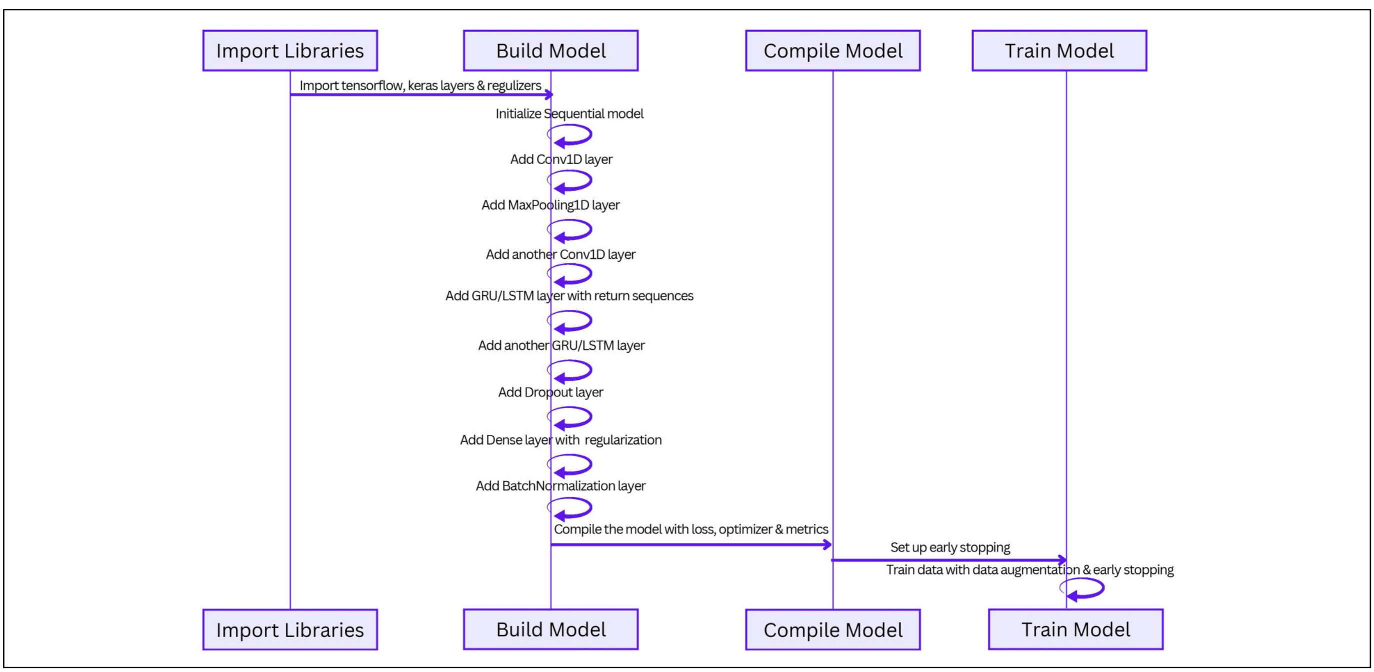

Figure 2. The architecture of the proposed deep learning models.

API components are imported to enable efficient model construction and regularization. The network is designed to process input data consisting of 125 time steps and 19 features, employing 1D Convolutional Layers with "relu" activation functions and specific filters to extract significant features from the input sequence. A Max Pooling Layer is integrated to reduce dimensionality while preserving essential features. In the 1DCNN-GRU model, 2 GRU layers were incorporated to capture temporal dependencies, with the first layer returning sequences and the second tailored to enhance overall model performance. Similarly, in the 1DCNN-LSTM model, LSTM layers were added to serve the same purpose. Additionally, a Dropout Layer is introduced for regularization purposes, mitigating overfitting risks. The architecture also incorporates a Dense Layer utilizing "softmax" activation and L2 regularization, complemented by Batch Normalization techniques to normalize inputs for subsequent layers.

The model is compiled using binary cross-entropy as the loss function and RMSprop as the optimizer, evaluating performance based on accuracy metrics. Early stopping, a callback mechanism, is implemented to prevent overfitting by halting training when there's no improvement in validation loss across a specified number of epochs.

Training involves the model being trained on the provided dataset, employing specific batch sizes and epochs while utilizing validation data for evaluating performance during training. This methodology ensures a robust and reliable model for the intended classification task.

Feature Importance

In our study, we used various feature importance methods to assess the contributions of individual features to the classification accuracy of our model. These methods allow us to evaluate the significance of each feature in predicting the target variable and gain insights into the relevance of each feature to the model.

Feature importance methods offer numerous advantages in data analysis by establishing clearer relationships between features and target variables. They help in identifying irrelevant features, reduce model dimensionality, enhance computational efficiency, and importantly, improve model interpretability by highlighting the most influential predictors for better predictive performance.

In our case, the features correspond to the electrodes used in our study, labeled as "Fp1," "Fp2," "F3," "F4," "C3," "C4," "P3," "P4," "O1," "O2," "F7," "F8," "T3," "T4," "T5," "T6," "Fz," "Cz," and "Pz."

The feature importances were calculated using Gini impurity, permutation, drop column, and XGBoost methods, respectively.[24–28] The most discriminative electrodes were selected after the comparative analysis.

Gini impurity: Each node in a decision tree specifies the conditions under which values in a single feature are to be split, ensuring that related values of the dependent variable are placed within the same set following the split. The criterion is based on impurity, which is variance for regression trees and Gini impurity/information gain for classification tasks. Consequently, one can calculate the contribution of each feature when training a tree.[24,25]

Permutation: Permutation feature importance estimates the feature importance by observing the way random reshuffling of every single predictor affects model performance. Training the baseline model, saving the score (accuracy/R2/any metric of interest), and afterward bypassing the validation set (or Out of Bag set when Random Forest is the case) are the essential phases in this algorithm. This is also possible on the training set but at the cost of sacrificing generalization-related data. First, the values need to be reshuffled from one feature in the selected dataset, then the dataset needs to be re-passed to the model to obtain predictions and evaluate the metrics for this updated dataset. The difference between the benchmark score and the score from the updated (permuted) dataset represents the feature importance. Summarizing, the feature's importance is evaluated highlighting the increase or decrease in error when the values of a feature are permuted. If permuting the values leads to a huge change in the error, it can be concluded that the feature is important for our model.[26,27]

Drop Column: The drop column importance method determines the degree to which a given variable improves, degrades or has no effect on a model's performance using a selected model performance metric.[29] To measure the significance of a feature, the method contrasts the model's performance with all features with a model that purposefully left out that feature during training.

XGBoost: XGBoost is an enhanced algorithm based on the gradient boosting decision tree that can efficiently build boosted trees and run in parallel.[28,30] The boosted trees within XGBoost are split into regression and classification trees, with the fundamental focus of the algorithm being the enhancement of the objective function's value through optimization. Unlike similarity calculations in traditional methods, XGBoost intelligently derives feature scores from the boosted trees, signifying each feature's importance to the model. As a feature plays a more pivotal role in decision-making within the boosted trees, its score correspondingly increases.

External Validation

External validation is a critical process in the field of machine learning, particularly in applications involving medical diagnostics and neurophysiological data interpretation. It refers to the evaluation of a trained machine learning model on a dataset that it has not encountered during the training phase. This contrasts with internal validation, where the model is evaluated on the same data used for training. External validation is essential for assessing the model's ability to generalize to new, unseen data, which is crucial in real-world applications where the data encountered in production may differ from the training data.

In this study, an external validation was performed on a distinct subset of data that was not part of the training phase. This subset consisted of 20 subjects equally divided between individuals with OCD and healthy controls. The model's performance was subsequently assessed on this independent test set to gauge its ability to generalize beyond the training data.

Results

The classification reports are represented for each model comparing precision, sensitivity, specificity, F1 score, test accuracy, and kappa value, respectively in Table 2. The information on the optimizer, loss function, number of epochs, and batch size are also included. According to the classification report, the precision (0.9707), sensitivity (0,9698), F1 score (0.9813), test accuracy (0.9811), and kappa value (0.9622) in the 1DCNN-LSTM model are higher than 1DCNN-GRU (precision = 0.9395; sensitivity = 0.9353; F1 score = 0.9683; test accuracy = 0.9672; kappa value = 0.9344). Whereas, the specificity of the 1DCNN-GRU (0.9988) was slightly higher than the specificity of the 1DCNN-LSTM (0.9922).

In Figure 3, the confusion matrices and ROC characteristics of the 2 models are compared. 1DCNN-LSTM model (ROC area = 0.98) gives a more robust and somewhat superior classification performance as compared to the 1DCNN-GRU model (ROC area = 0.97). Figure 4 summarizes the history of the classification accuracy and loss functions for each training epoch.

Cross-Validation and External Validation Results

The 5-fold cross-validation results are given in terms of the mean and standard deviation values in Table 3. The mean classification accuracies in the cross-validation stage were 90.88% (±1.25) and 85.91% (±2.45) for the 1DCNN-LSTM and 1DCNN-GRU models, respectively.

Given the superior performance of the 1DCNN-LSTM model, it underwent external validation. In Supplement 1, the

Table 2. Classification Report.

	Precision	Sensitivity	Specificity	F1 score	Test accuracy
1DCNN-LSTM	0.9707	0.9698	0.9922	0.9813	0.9811
1DCNN-GRU	0.9395	0.9353	0.9988	0.9683	0.9672
	Kappa value	Optimizer	Loss function	Number of epochs	Batch size
1DCNN-LSTM	0.9622	adam	binary_crossentropy	150	32
1DCNN-GRU	0.9344	adam	binary_crossentropy	50	128

Abbreviations: 1DCNN, one-dimensional convolutional neural networks; GRU, gradient recurrent units; LSTM, long-short term memory.

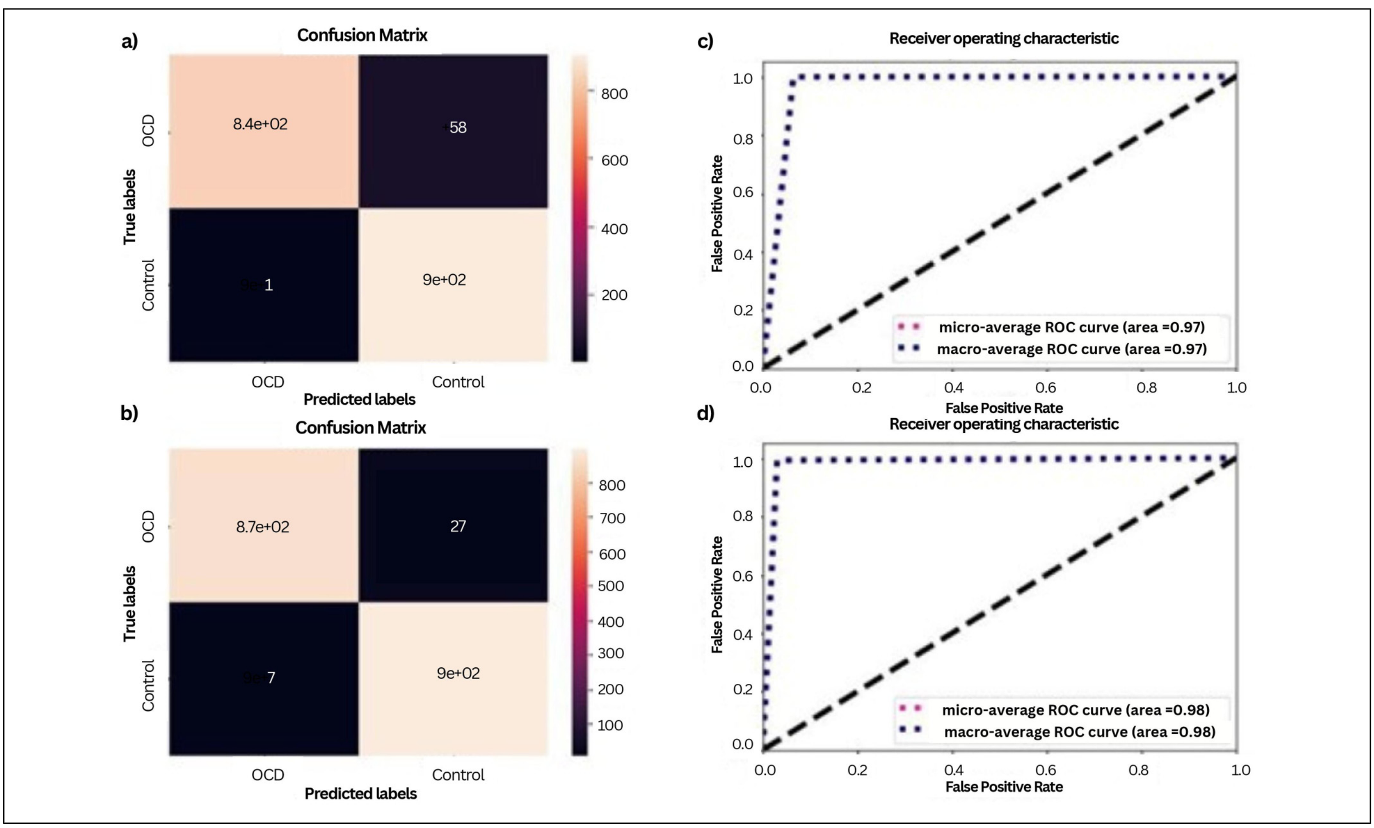

Figure 3. (a) Confusion matrix of 1DCNN-GRU; (b) confusion matrix of 1DCNN-LSTM; (c) ROC characteristics of GRU; and (d) ROC characteristics of LSTM.
Abbreviations: 1DCNN, one-dimensional convolutional neural networks; GRU, gradient recurrent units; LSTM, long-short term memory; ROC, receiver operating characteristic.

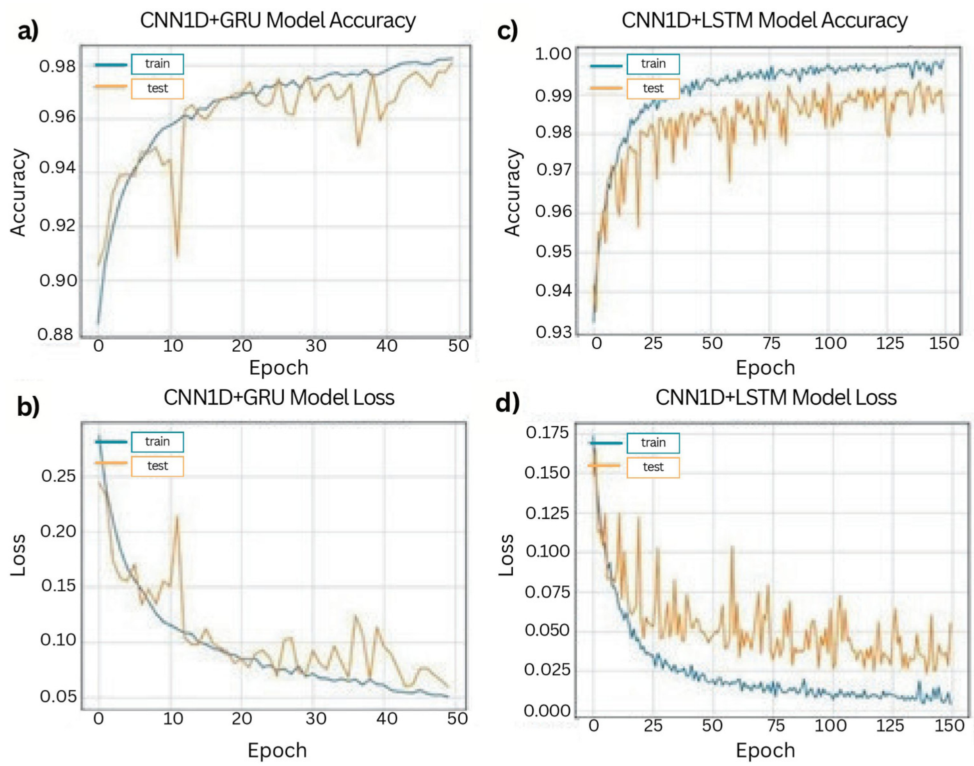

Figure 4. Summarize history for (a) accuracy and (b) loss function of 1DCNN-GRU and accuracy (c) and loss (d) function of 1DCNN-LSTM. Abbreviations: 1DCNN, one-dimensional convolutional neural networks; GRU, gradient recurrent units; LSTM, long-short term memory.

Table 3. Cross-Validated Results of the Proposed Models.

1DCNN + GRU		1DCNN + LSTM	
	Accuracy (%)		Accuracy (%)
Fold 1	86.17	Fold 1	91.94
Fold 2	88.11	Fold 2	91.89
Fold 3	85.72	Fold 3	89.61
Fold 4	81.89	Fold 4	91.56
Fold 5	87.67	Fold 5	89.44
Mean	85.91	Mean	90.88
Standard deviation	2,45	Standard deviation	1,25

Abbreviations: 1DCNN, one-dimensional convolutional neural networks; GRU, gradient recurrent units; LSTM, long-short term memory.

confusion matrix of the external validation provides a clear visualization of the model's performance, showing the number of true positives, true negatives, false positives, and false negatives.

The classification report for the external validation is presented in Supplement 2, indicating metrics as specificity (0.80), sensitivity (0.70), test accuracy (0.75), F1 score (0.74), and kappa value (0.50) for the model.

Feature Importance Results

The feature importances have been identified, and significant electrodes are marked on the topography presented in Figure 5. Based on the results, the top 4 most important features for each feature importance method are given in Table 4, and the importances of each electrode according to the 4 different feature importance techniques are represented in Supplement 3. In the Gini impurity feature extraction, the most discriminative channels were O2, T5, T6, and F7. The Permutation method highlighted T5, O2, PT, and F7 as important channels. When using the Drop column method, significant channels were found to be O2, F7, T6, and T4. In the XGBoost method, T6, O2, T5, and F7 emerged as the key channels. Although the electrodes that contribute most to the discrimination differ between methods, we have observed dominance of inferior frontal, temporal, and occipital electrodes.

Discussion

In this study, we applied deep learning method to classify OCD EEG data and tested the discrimination accuracy of LSTM and GRU architectures.

CNNs are designed to process data with a grid-like topology, such as images, and are composed of multiple layers of neurons

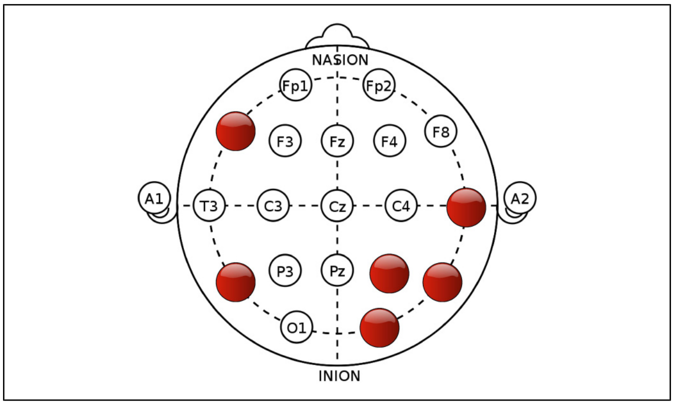

Figure 5. Topographical representation of discriminative channels. In the figure, electrodes F7, T5, P4, T4, T6, and O2 are highlighted to illustrate the discriminative features.

Table 4. The Most Discriminative Features With Respect to Various Feature Importance Methods.

Method	Discriminative features
Gini impurity	O2-T5-T6-F7
Permutation	T5-O2-P4-F7
Drop column	O2-F7-T6-T4
XGBoost	T6-O2-T5-F7

that apply filters to the input data to extract features that are relevant to the task at hand. LSTM network, on the other hand, is a type of RNN that is designed to process sequential data, such as time series or natural language text. LSTMs use a set of gates to control the flow of information through the network, allowing them to capture long-term dependencies in the data. LSTMs, characterized by 3 gates (input, forget, and output), offer refined memory control, enabling them to capture intricate temporal dependencies. On the other hand, GRUs are another type of RNN that are similar to LSTMs, but they have a simpler structure and are typically easier to train. Like LSTMs, GRUs use gates to control the flow of information through the network, but they use a different set of equations to update the hidden state of the network. GRUs possess a simpler structure with 2 gates (reset and update), resulting in a more compact model with fewer parameters.

Combining CNNs and RNNs, as in the case of CNN-LSTM and CNN-GRU models, can allow for the extraction of both spatial and temporal features from the data, leading to improved performance on many tasks. By integrating 1DCNN with LSTM or GRU, the study aims to leverage the strengths of CNNs in spatial feature extraction with the temporal processing capabilities of RNN variants, aiming to provide a more comprehensive analysis.

Both models showed excellent discrimination accuracy with notably higher classification accuracy for the 1DCNN-LSTM model as compared to the 1DCNN-GRU model. The computational efficiency of the GRU model was a primary consideration due to its faster training times. However, despite the GRU's efficiency, the LSTM's more intricate structure, which potentially allows for more refined memory control, resulted in superior performance in classifying OCD EEG signals.

It was also found that while the 1DCNN-LSTM model had a clear advantage in several metrics like accuracy, sensitivity, and F1 score, the GRU model had a slight advantage in specificity. The GRU's slightly higher specificity could be attributed to its architectural simplicity relative to LSTM. GRU cells have a more compact structure with fewer parameters and less intricate gating mechanisms compared to LSTM. This streamlined architecture might enable the GRU model to focus more sharply on specific features or patterns relevant to true negatives. However, during the external validation phase, LSTM's performance decreased, underscoring the importance of continuous refinement and validation of machine learning models, especially in the context of medical diagnostics and neuroscientific research.

Different feature extraction methods highlighted discriminating channels, showing consistency in the significance of electrodes in the inferior frontal, temporal, and right occipital regions. This consensus across different techniques suggests a potential agreement on the crucial role of these brain areas in feature discrimination.

Previous neuroimaging studies showed somewhat contradicting results as some reported mainly frontostriatal alterations,[31] while others mainly found temporoparietal alterations.[4] Our findings in terms of feature importance suggest that both frontals and limbic alterations in addition to abnormalities related to primary sensory areas such as the visual cortex might be involved in OCD. Thus a frontostriatal model of OCD[32] could be refuted and a new model including somatosensorial areas could be proposed. Such models are also supported by studies reporting bottom-up sensory alterations in OCD[33] and based on such studies one might argue that the top-down and bottom-up pathways are both involved in OCD.

Our work expands beyond a novel algorithmic approach; it evaluates a hybrid deep learning method tailored for OCD classification using EEG data, which may be more effective at detecting subtle patterns in the data than classical signal processing and feature extraction methods. Beyond classification, our approach aims to extract biomarkers, unraveling the underlying mechanisms of the disorder. The results imply that OCD EEG includes very distinctive features that can be revealed with sophisticated deep learning methods and that are not recognizable using classical signal processing and feature extraction methods that were used in previous studies. Additionally, the results of the study suggest that both frontostriatal and temporoparietal alterations may be involved in OCD, which could support the development of new models of the disorder that incorporate both executive and somatosensorial areas.

For better generalizability, we plan to investigate our models and compare their performances using larger datasets in our future works.

Acknowledgments

We are grateful to Dr Elvan Çiftçi for her contribution to the revision process and the NP Istanbul Brain Hospital for granting access to their database, which made this research possible.

Data Availability Statement

Due to privacy regulations, the dataset analyzed during the current work is not publicly accessible, but it is available from the corresponding author upon justifiable request.

Declaration of Conflicting Interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The authors received no financial support for the research, authorship, and/or publication of this article.

Ethical Statement

The access to the database and subsequent data analysis were granted ethical approval by the Uskudar University Ethics Committee under approval number E-11111119-307.99, affirming adherence to ethical standards, participant confidentiality, and regulatory compliance.

ORCID iDs

Shams Farhad https://orcid.org/0000-0003-0591-2765
Sinem Zeynep Metin https://orcid.org/0000-0002-0511-2867
Barış Metin https://orcid.org/0000-0003-1626-058X
Türker Tekin Ergüzel https://orcid.org/0000-0001-8438-6542

Supplemental Material

Supplemental material for this article is available online.

References

1. Ruscio AM, Stein DJ, Chiu WT, Kessler RC. The epidemiology of obsessive-compulsive disorder in the national comorbidity survey replication. *Mol Psychiatry*. 2008;15(1):53-63. doi:10.1038/mp.2008.94
2. Clemente MJ, Martins Silva AS, Pozzolo Pedro MO, et al. A meta-analysis and meta-regression analysis of the global prevalence of obsessive-compulsive personality disorder. *Heliyon*. 2022;8(7). doi:10.1016/j.heliyon.2022.e09912
3. Sharma E, Sharma LP, Balachander S, et al. Comorbidities in obsessive-compulsive disorder across the lifespan: a systematic review and meta-analysis. *Front Psychiatry*. 2021:12. doi:10.3389/fpsyt.2021.703701
4. Boedhoe PSW, Schmaal L, Abe Y, et al. Cortical abnormalities associated with pediatric and adult obsessive-compulsive disorder: findings from the ENIGMA obsessive-compulsive disorder working group. *Am J Psychiatry*. 2018;175(5):453-462. doi:10.1176/appi.ajp.2017.17050485
5. Bruin WB, Taylor L, Thomas RM, et al. Structural neuroimaging biomarkers for obsessive-compulsive disorder in the ENIGMA-OCD consortium: medication matters. *Transl Psychiatry*. 2020;10(1):342. Published 2020 Oct 8. doi:10.1038/s41398-020-01013-y
6. Zhan Y, Wei J, Liang J, et al. Diagnostic classification for human autism and obsessive-compulsive disorder based on machine learning from a primate genetic model. *Am J Psychiatry*. 2021;178(1):65-76. doi:10.1176/appi.ajp.2020.19101091
7. Smith EE, Schüller T, Huys D, et al. A brief demonstration of frontostriatal connectivity in OCD patients with intracranial electrodes. *Neuroimage*. 2020;220:117138. doi:10.1016/j.neuroimage.2020.117138
8. Cheng J, Li P, Tang Y, et al. Transcranial direct current stimulation improve symptoms and modulates cortical inhibition in obsessive-compulsive disorder: a TMS-EEG study. *J Affect Disord*. 2022;298(Pt A):558-564. doi:10.1016/j.jad.2021.11.043
9. Metin SZ, Balli Altuglu T, Metin B, et al. Use of EEG for predicting treatment response to transcranial magnetic stimulation in obsessive compulsive disorder. *Clin EEG Neurosci*. 2019;51(3):139-145. doi:10.1177/1550059419879569
10. Altuğlu TB, Metin B, Tülay EE, et al. Prediction of treatment resistance in obsessive compulsive disorder patients based on EEG complexity as a biomarker. *Clin Neurophysiol*. 2020;131(3):716-724. doi:10.1016/j.clinph.2019.11.063
11. Balci F, Oralhan Z. LSTM Ile EEG Tabanlı kimliklendirme sistemi Tasarımı. *Eur J Sci Technol*. Published online 2020:135-141. doi:10.31590/ejosat.779526
12. Genis Y, Akman Aydin E. Diagnosis of epilepsy disease with deep learning methods using EEG signals. 2022 30th Signal Processing and Communications Applications Conference (SIU). Published online 2022. doi:10.1109/siu55565.2022.9864765
13. Saeedi A, Saeedi M, Maghsoudi A, Shalbaf A. Major depressive disorder diagnosis based on effective connectivity in EEG signals: a convolutional neural network and long short-term memory approach. *Cogn Neurodyn*. 2020;15(2):239-252. doi:10.1007/s11571-020-09619-0
14. Mumtaz W, Qayyum A. A deep learning framework for automatic diagnosis of unipolar depression. *Int J Med Inf*. 2019;132:103983. doi:10.1016/j.ijmedinf.2019.103983
15. Xu G, Ren T, Chen Y, Che W. A one-dimensional CNN-LSTM model for epileptic seizure recognition using EEG signal analysis. *Front Neurosci*. 2020:14. doi:10.3389/fnins.2020.578126
16. Gao S, Yang J, Shen T, Jiang W. A parallel feature fusion network combining GRU and CNN for motor imagery EEG decoding. *Brain Sci*. 2022;12(9):1233. doi:10.3390/brainsci12091233
17. Qiao W, Bi X. Deep Spatial-temporal neural network for classification of EEG-based motor imagery. Proceedings of the 2019 International Conference on Artificial Intelligence and Computer Science. Published online 2019. doi:10.1145/3349341.3349414
18. Isuru Niroshana SM, Zhu X, Chen Y, Chen W. Sleep stage classification based on EEG, EOG, and CNN-GRU deep learning model. 2019 IEEE 10th International Conference on Awareness Science and Technology (iCAST). Published online 2019. doi:10.1109/icawst.2019.8923359
19. Zou G. The recognition of action idea EEG with deep learning. *Complexity*. 2022;2022:1-13. doi:10.1155/2022/5308885
20. Zou GZ. CONV1D and GRU-based EEG emotion recognition method. International Conference on Computer Graphics, Artificial Intelligence, and Data Processing (ICCAID 2021). Published online 2022. doi:10.1117/12.2631432
21. Zhao H. Target EEG recognition based on CNN and bidirectional GRU hybrid model. International Conference on Image, Signal Processing, and Pattern Recognition (ISPP 2022). Published online 2022. doi:10.1117/12.2636802

22. Menaka R, Karthik R, Saranya S, Niranjan M, Kabilan S. An improved AlexNet model and cepstral coefficient-based classification of autism using EEG [published online ahead of print, 2023 May 29]. *Clin EEG Neurosci*. 2023:15500594231178274. doi:10.1177/15500594231178274
23. Metin B, Uyulan Ç, Ergüzel TT, et al. The deep learning method differentiates patients with bipolar disorder from controls with high accuracy using EEG data. *Clin EEG Neurosci*. Published online 2022:155005942211372. doi:10.1177/15500594221137234
24. Menze BH, Kelm BM, Masuch R, et al. A comparison of random forest and its Gini importance with standard chemometric methods for the feature selection and classification of Spectral Data. *BMC Bioinformatics*. 2009;10(1). doi:10.1186/1471-2105-10-213
25. Qi Y. Random forest for bioinformatics. *Ensemble Machine Learning*. Published online 2012:307-323. doi:10.1007/978-1-4419-9326-7_11
26. Altmann A, Toloşi L, Sander O, Lengauer T. Permutation importance: a corrected feature importance measure. *Bioinformatics*. 2010;26(10):1340-1347. doi:10.1093/bioinformatics/btq134
27. Kaneko H. Cross-validated permutation feature importance considering correlation between features. *Analyt Sci Adv*. 2022; 3(9-10):278-287. doi:10.1002/ansa.202200018
28. Zheng H, Yuan J, Chen L. Short-term load forecasting using EMD-LSTM neural networks with a Xgboost algorithm for feature importance evaluation. *Energies*. 2017;10(8):1168. doi:10.3390/en10081168
29. Chen C, Liaw A, Breiman L. Using random forest to learn imbalanced data. *Univ Calif, Berkeley*. 2004;110(1-12):24.
30. Friedman JH. Stochastic gradient boosting. *Comput Stat Data Anal*. 2002;38(4):367-378.
31. Hu X, Du M, Chen L, et al. Meta-analytic investigations of common and distinct grey matter alterations in youths and adults with obsessive-compulsive disorder. *Neurosci Biobehav Rev*. 2017;78:91-103. doi:10.1016/j.neubiorev.2017.04.012
32. Melloni M, Urbistondo C, Sedeño L, Gelormini C, Kichic R, Ibanez A. The extended fronto-striatal model of obsessive compulsive disorder: convergence from event-related potentials, neuropsychology and neuroimaging. *Front Hum Neurosci*. 2012;6. doi:10.3389/fnhum.2012.00259
33. Poletti M, Gebhardt E, Pelizza L, Preti A, Raballo A. Neurodevelopmental antecedents and sensory phenomena in obsessive compulsive disorder: a systematic review supporting a phenomenological-developmental model. *Psychopathology*. 2022;56(4):295-305. doi:10.1159/000526708